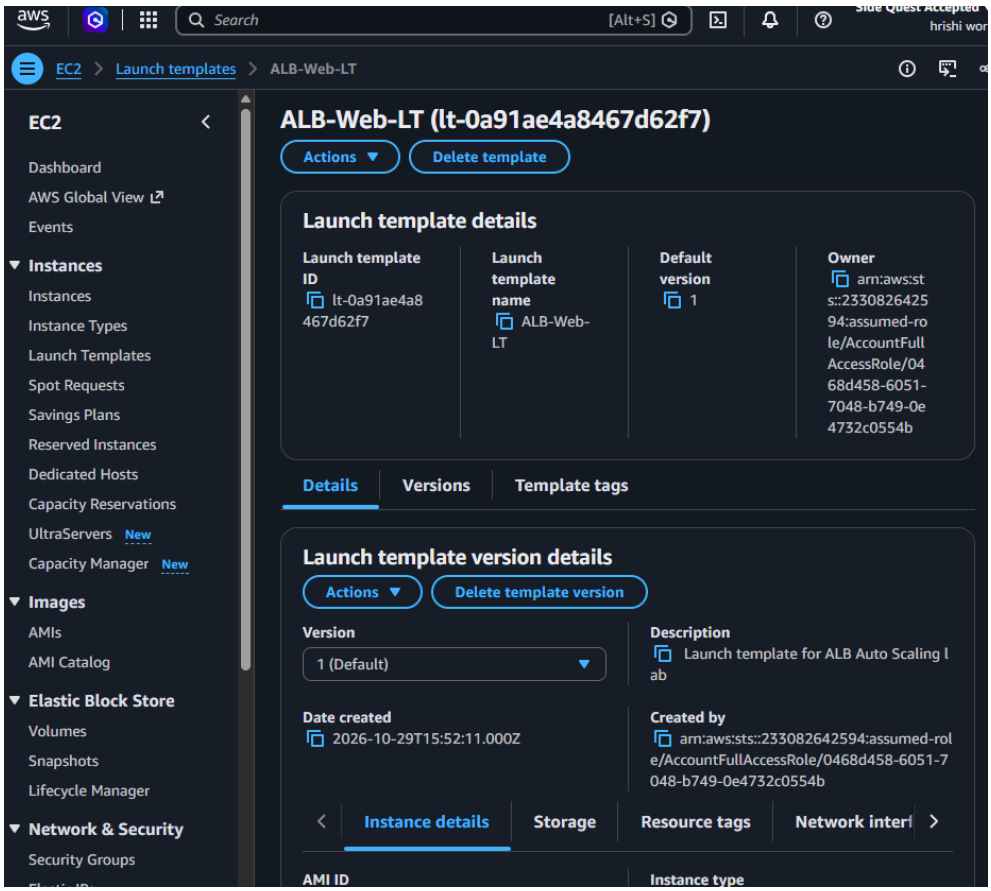
AWS EC2 Auto Scaling Lab

Submitted by Hrishinandan P

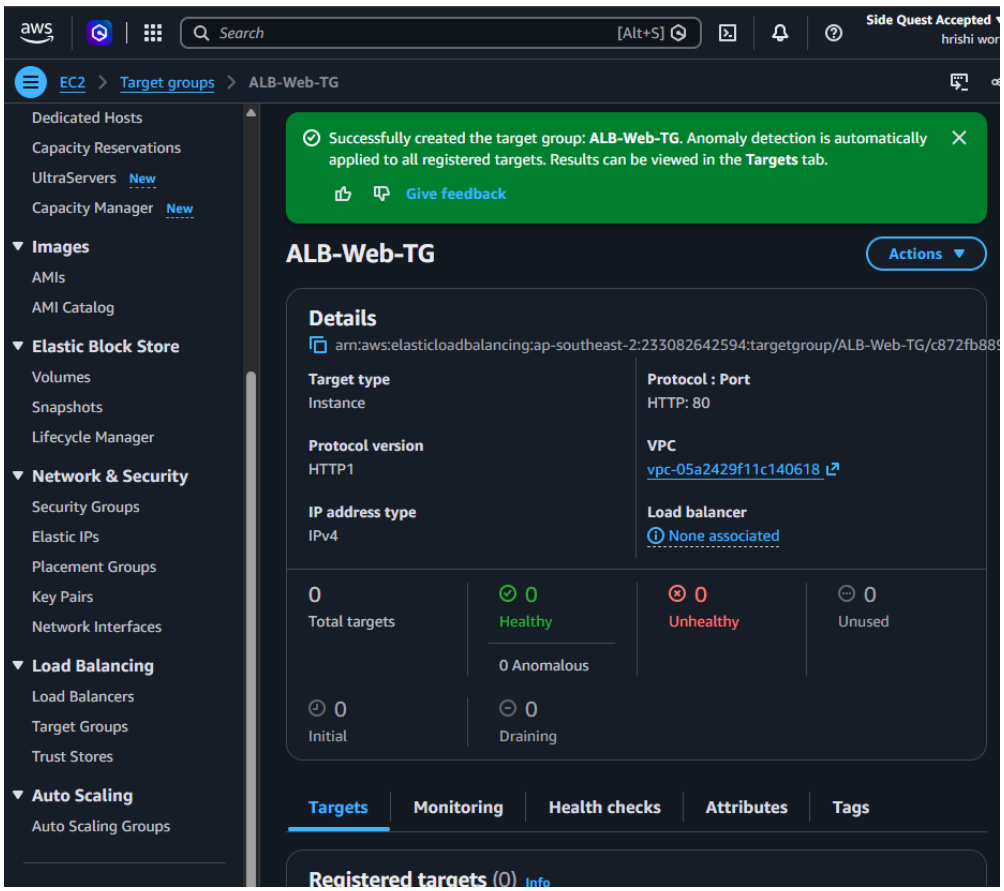
Task 1: Create a Launch Template

- Created launch template *ALB-Web-LT* with Amazon Linux, SSH/HTTP security group, key pair, and user data to install Apache and display the hostname.
- Version 1 is set as the default version.



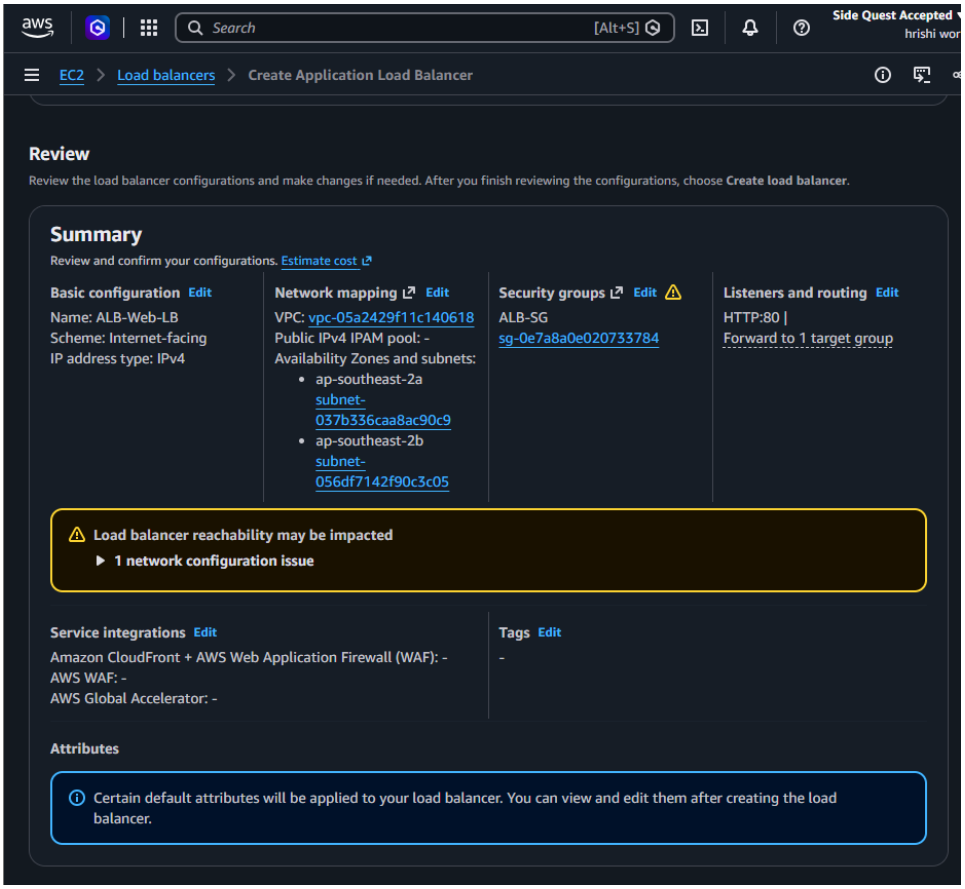
Task 2: Create a Target Group

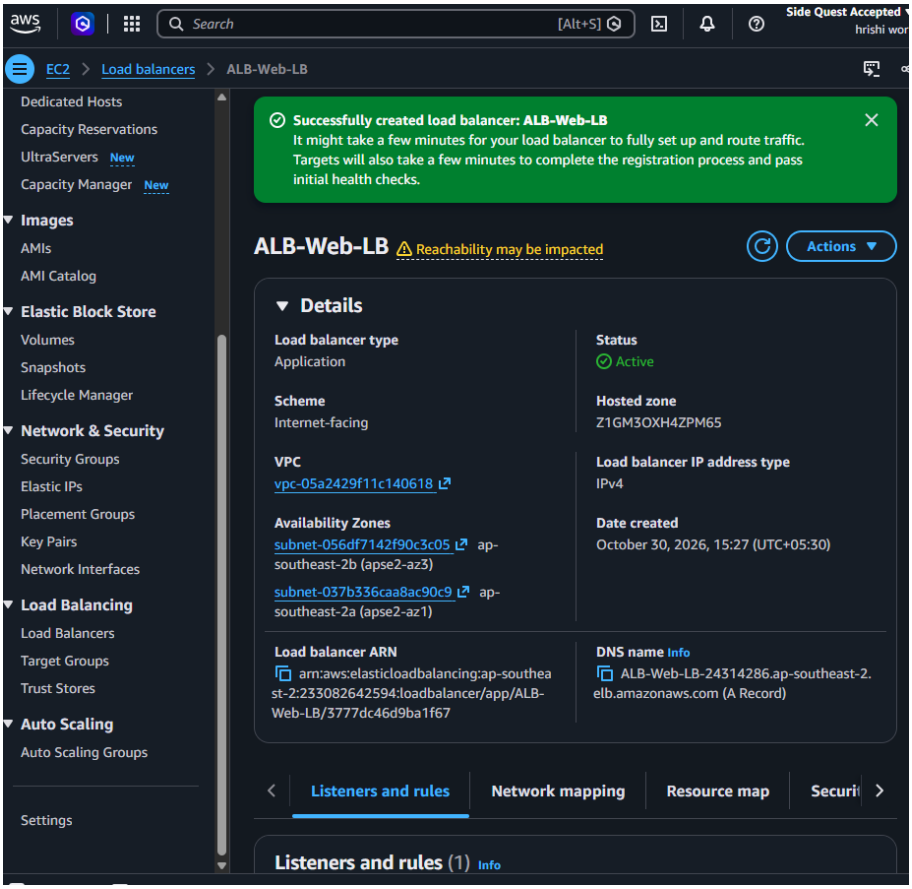
- Created instance target group *ALB-Web-TG* (HTTP: 80) with HTTP health checks on */*.
- No instances were registered manually (0 targets); the Auto Scaling Group registers them automatically.



Task 3: Create an Application Load Balancer

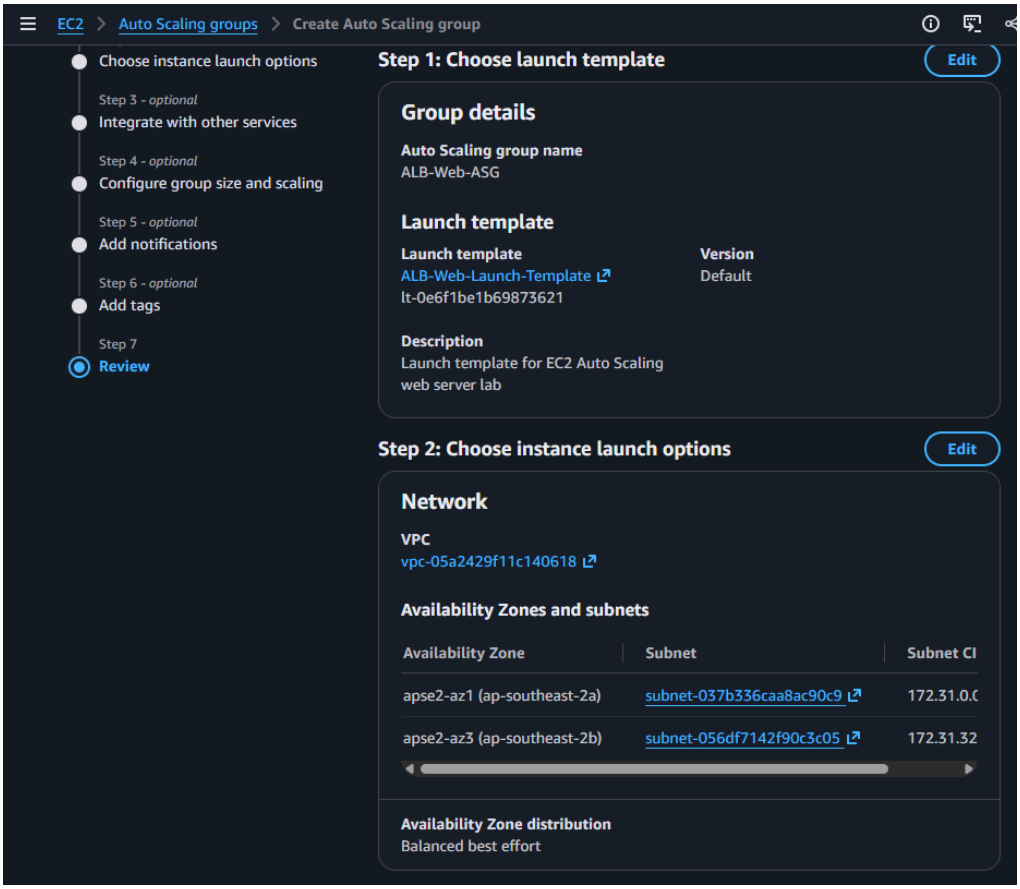
- Created internet-facing ALB *ALB-Web-LB* across two public subnets (ap-southeast-2a and 2b), with an HTTP:80 listener forwarding to *ALB-Web-TG*.
- The load balancer is Active and has a DNS name for access.

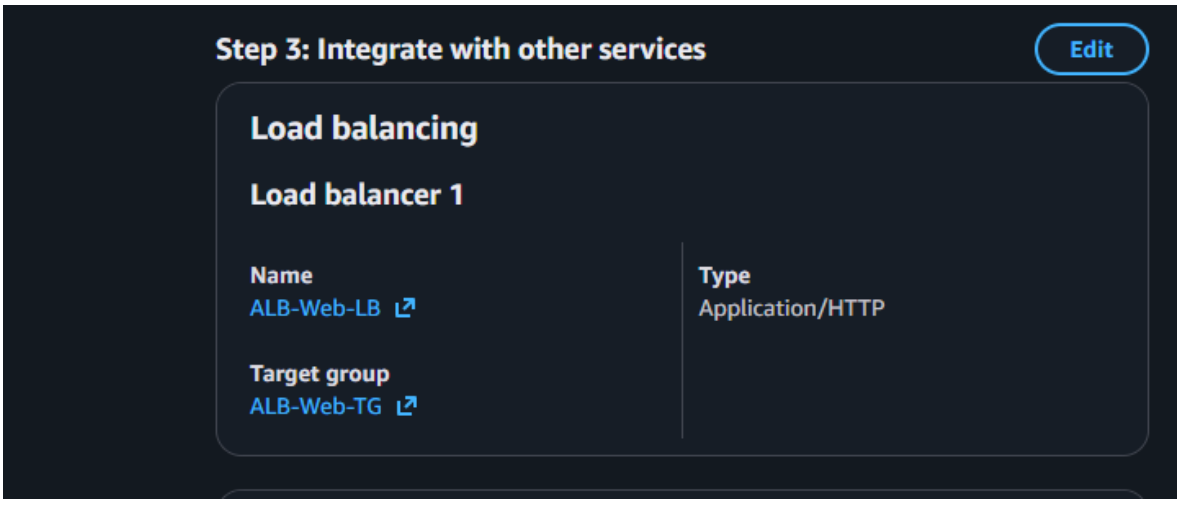



Search
[Alt+S]
Side Quest Accepted
hrishi wor
EC2
Load balancers
ALB-Web-LB
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Auto Scaling Groups
Settings
Successfully created load balancer: ALB-Web-LB
It might take a few minutes for your load balancer to fully set up and route traffic.
Targets will also take a few minutes to complete the registration process and pass initial health checks.
ALB-Web-LB
Reachability may be impacted
Actions
Details
Load balancer type
Application
Status
Active
Scheme
Internet-facing
Hosted zone
Z1GM3OXH4ZPM65
VPC
vpc-05a2429f11c140618
Load balancer IP address type
IPv4
Availability Zones
subnet-056df7142f90c3c05 ap-southeast-2b (apse2-az3)
subnet-037b336caa8ac90c9 ap-southeast-2a (apse2-az1)
Date created
October 30, 2026, 15:27 (UTC+05:30)
Load balancer ARN
arn:aws:elasticloadbalancing:ap-southeast-2:233082642594:loadbalancer/app/ALB-Web-LB/3777dc46d9ba1f67
DNS name Info
ALB-Web-LB-24314286.ap-southeast-2.elb.amazonaws.com (A Record)
Listeners and rules
Network mapping
Resource map
Securi
Listeners and rules (1) Info

Task 4: Create an Auto Scaling Group

- Created *ALB-Web-ASG* using the launch template, attached to target group *ALB-Web-TG*, across two public subnets.
- Configured Min=2, Desired=2, Max=5; the group is at desired capacity.





Step 4: Configure group size and scaling policies

Edit

Group size

Desired capacity
2

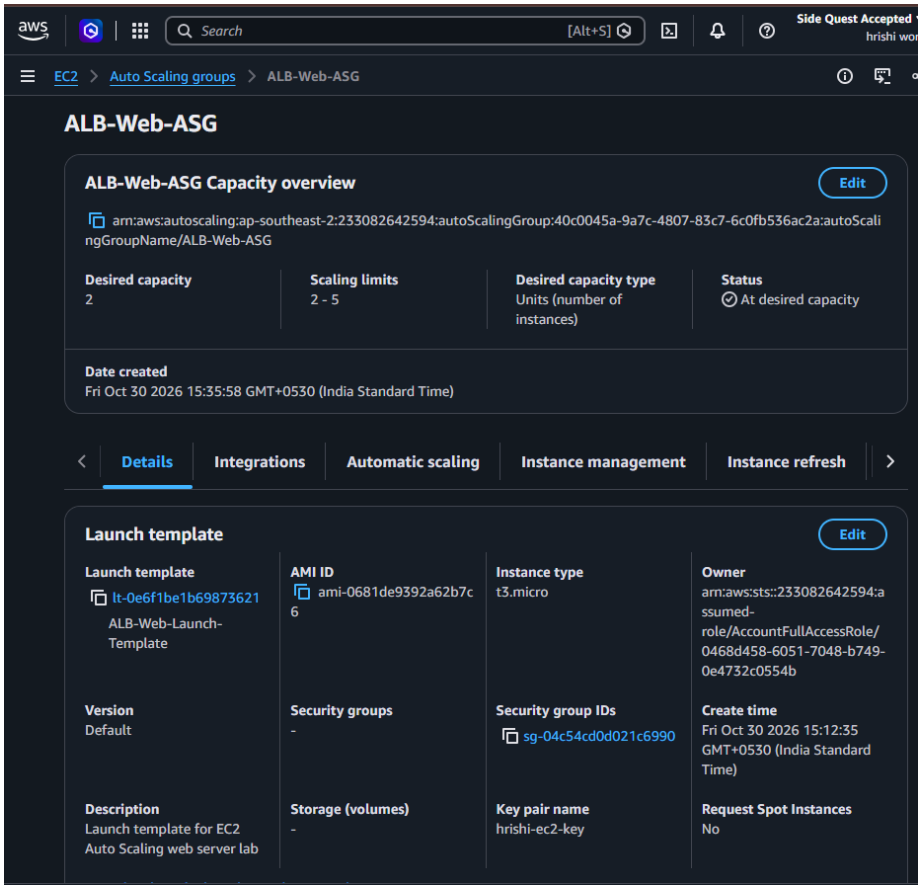
Desired capacity type
Units (number of instances)

Scaling

Minimum desired capacity
2

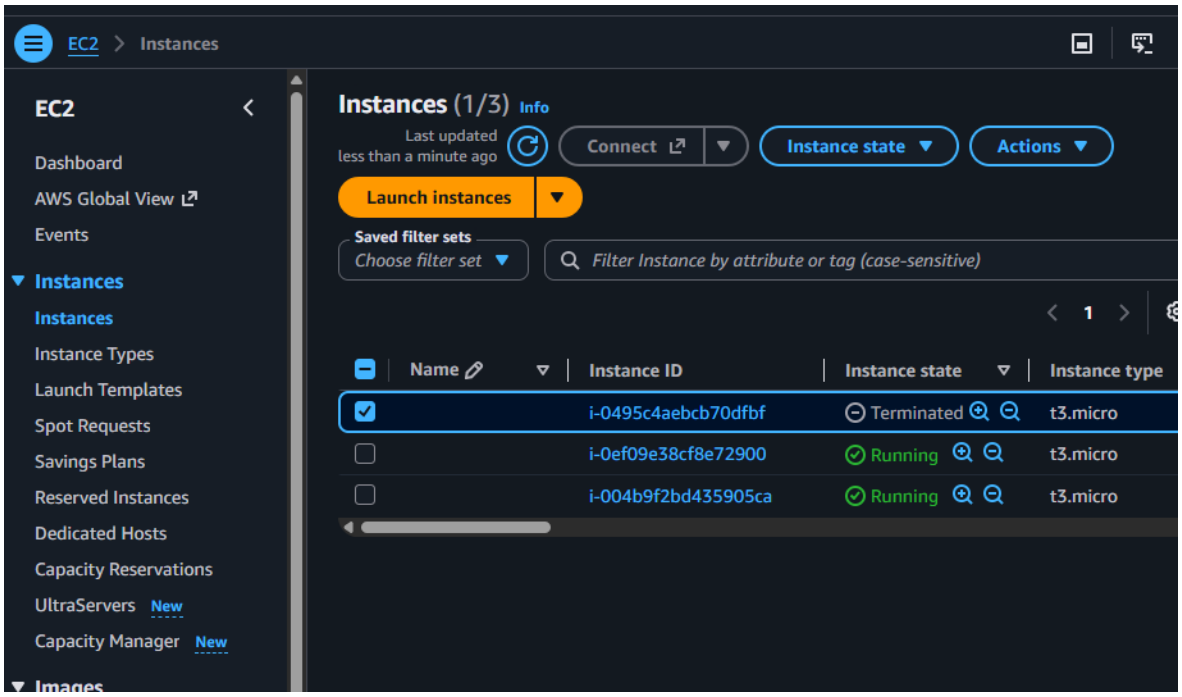
Maximum desired capacity
5

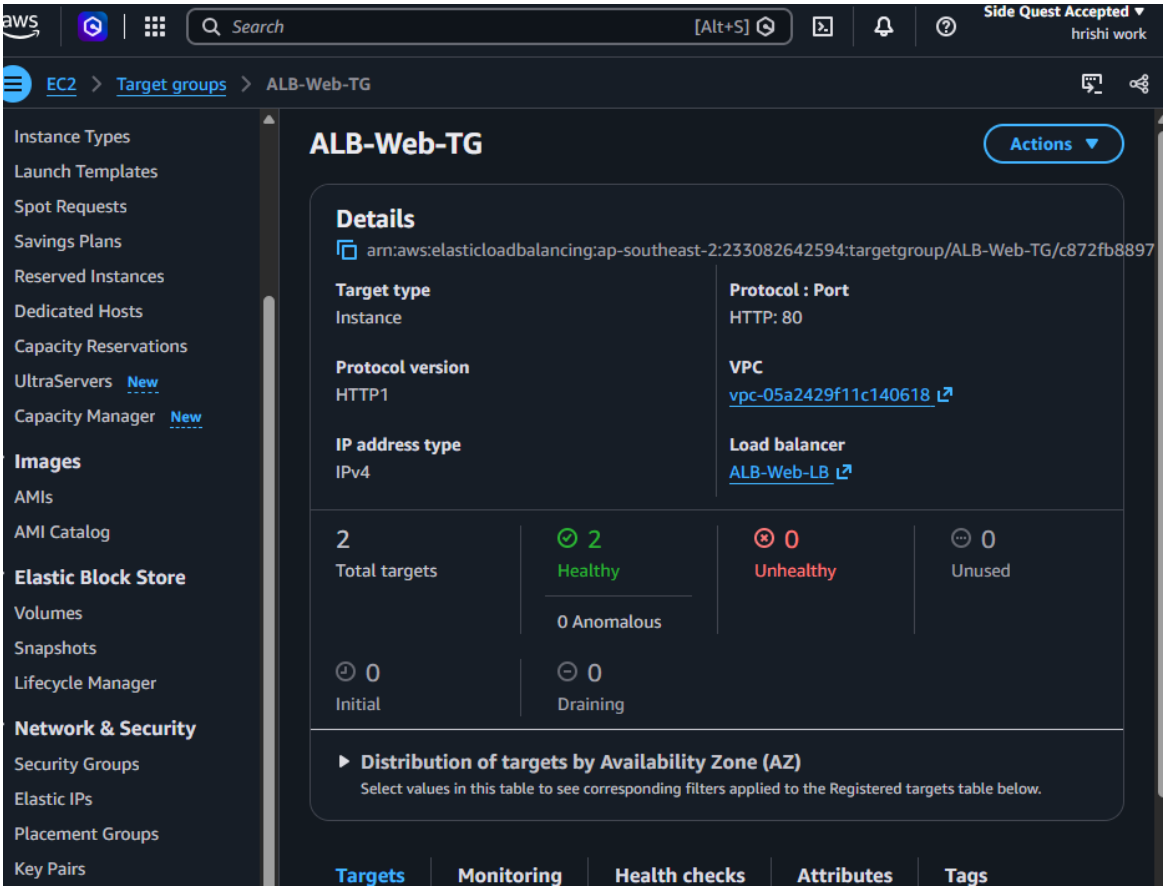
Target tracking policy
-



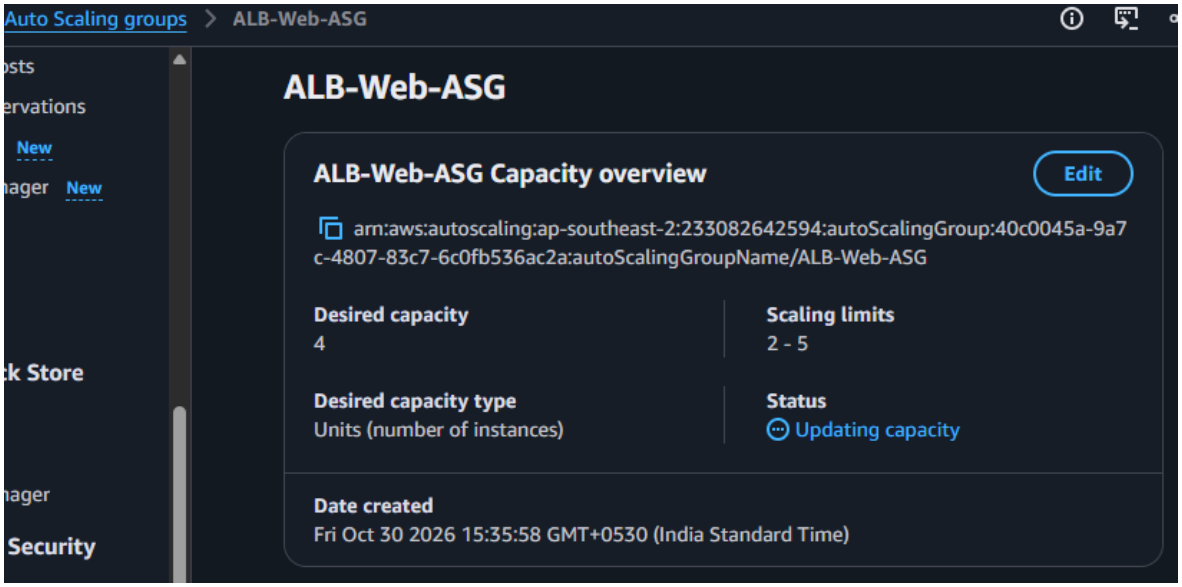
Task 5: Manual Scaling

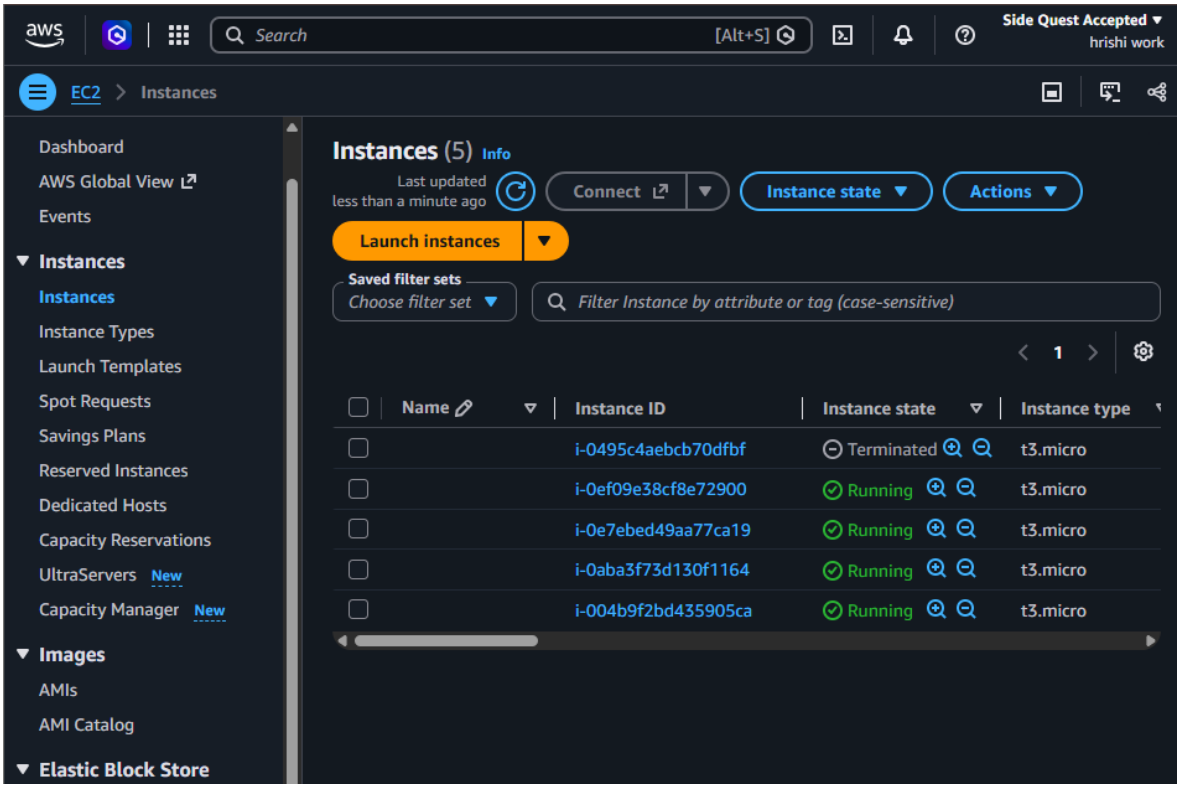
- Increased desired capacity from 2 to 4; two new instances launched and all 4 targets became healthy.
- Reduced desired capacity back to 2; the extra instances were drained and terminated, leaving 2 running.

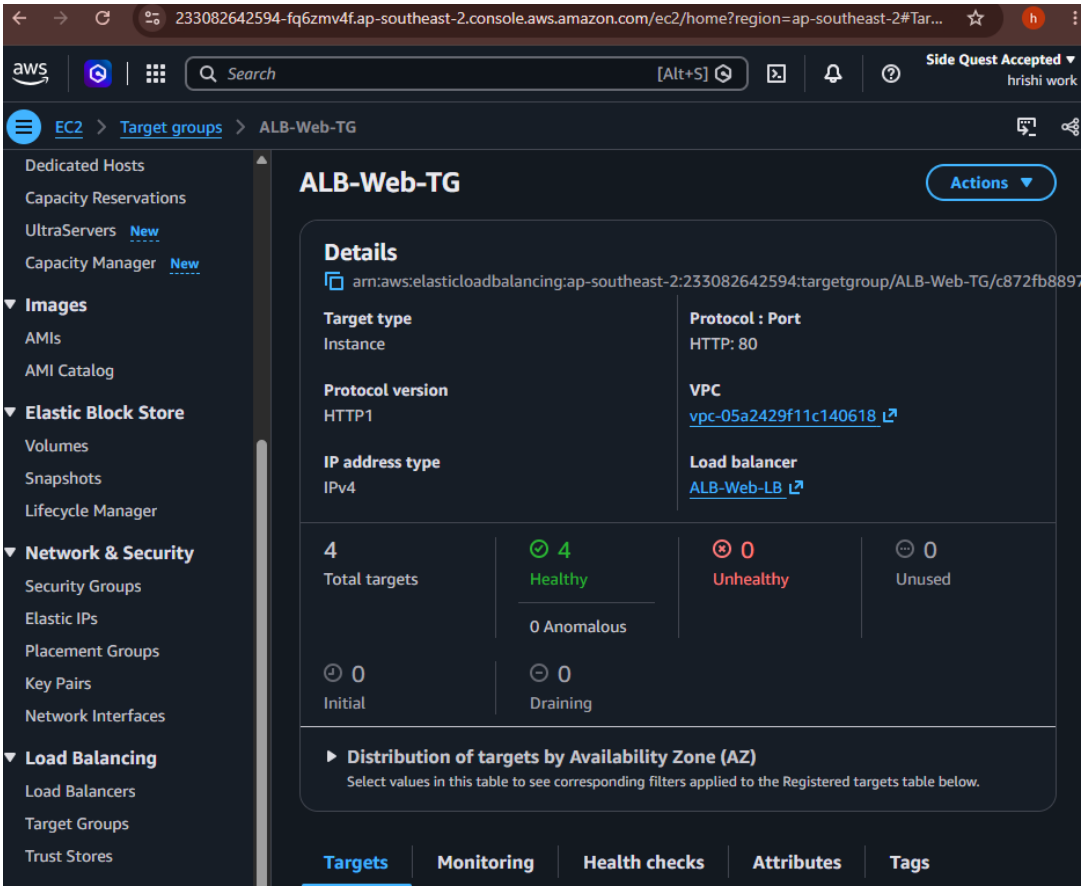


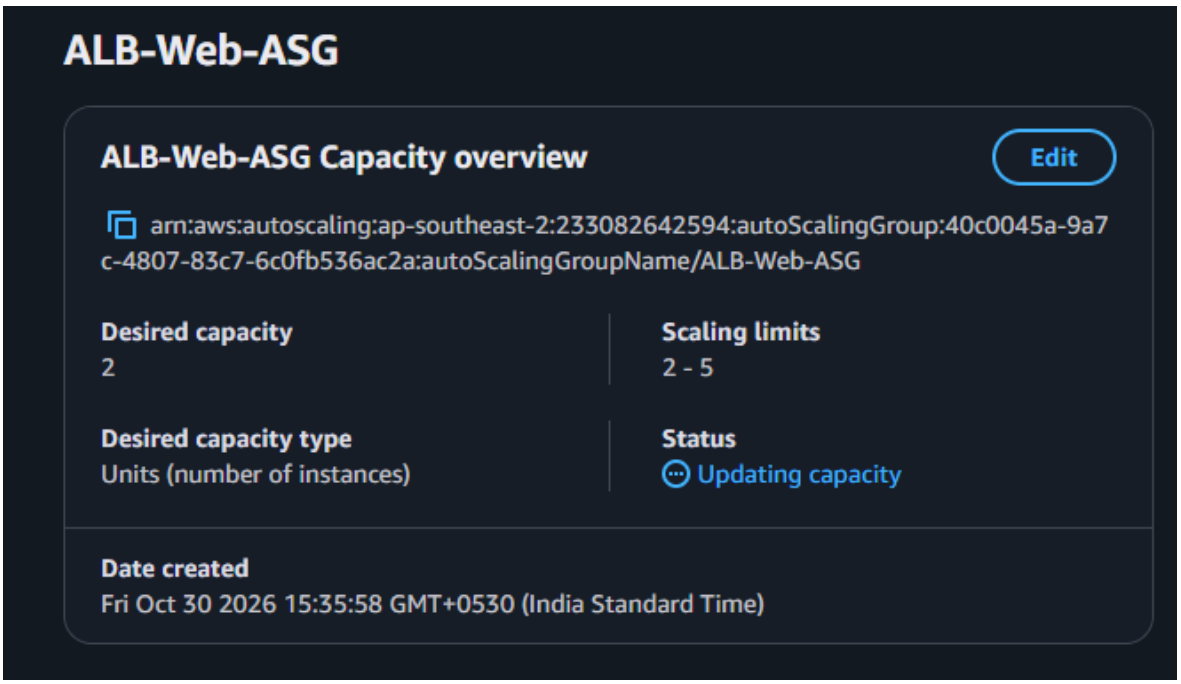


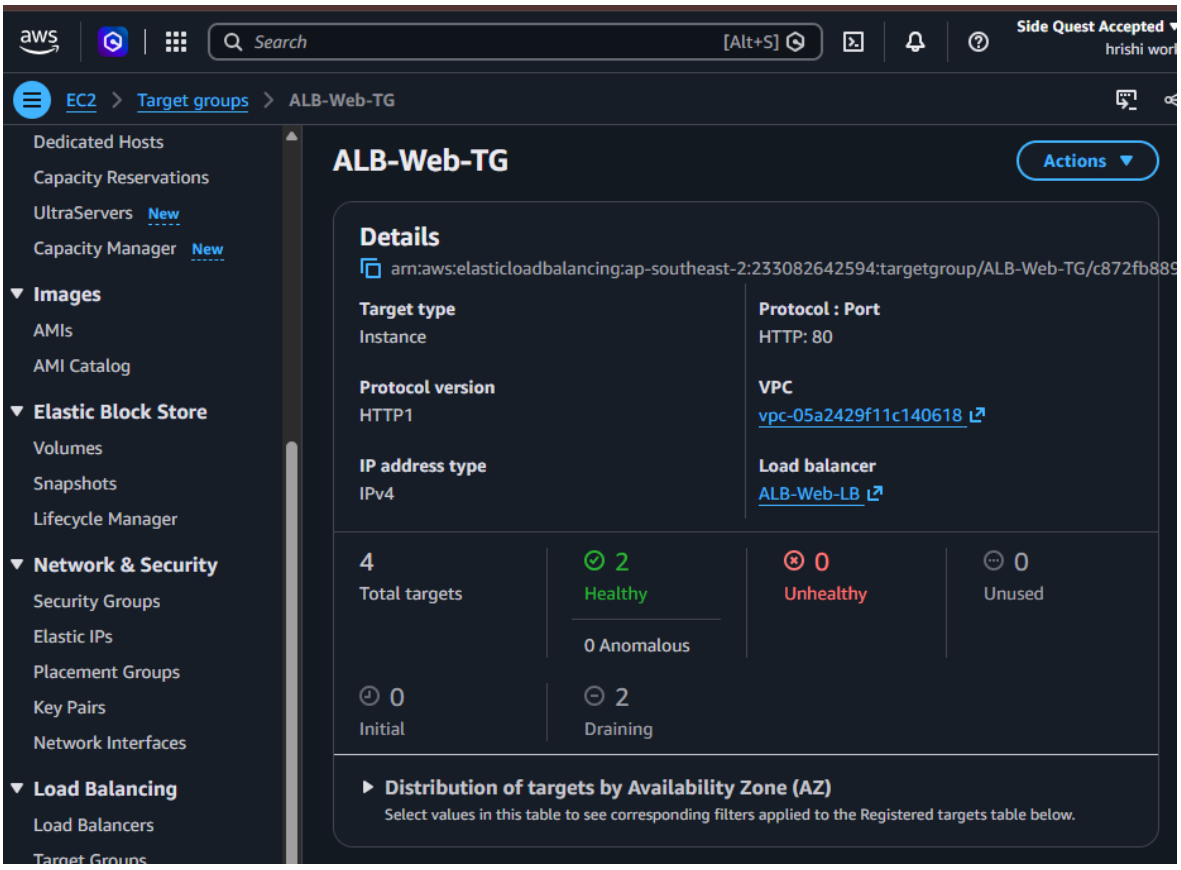

Auto Scaling groups
ALB-Web-ASG
ALB-Web-ASG
ALB-Web-ASG Capacity overview
Edit
arn:aws:autoscaling:ap-southeast-2:233082642594:autoScalingGroup:40c0045a-9a7c-4807-83c7-6c0fb536ac2a:autoScalingGroupName/ALB-Web-ASG
Desired capacity
2
Scaling limits
2 - 5
Desired capacity type
Units (number of instances)
Status
At desired capacity
Date created
Fri Oct 30 2026 15:35:58 GMT+0530 (India Standard Time)

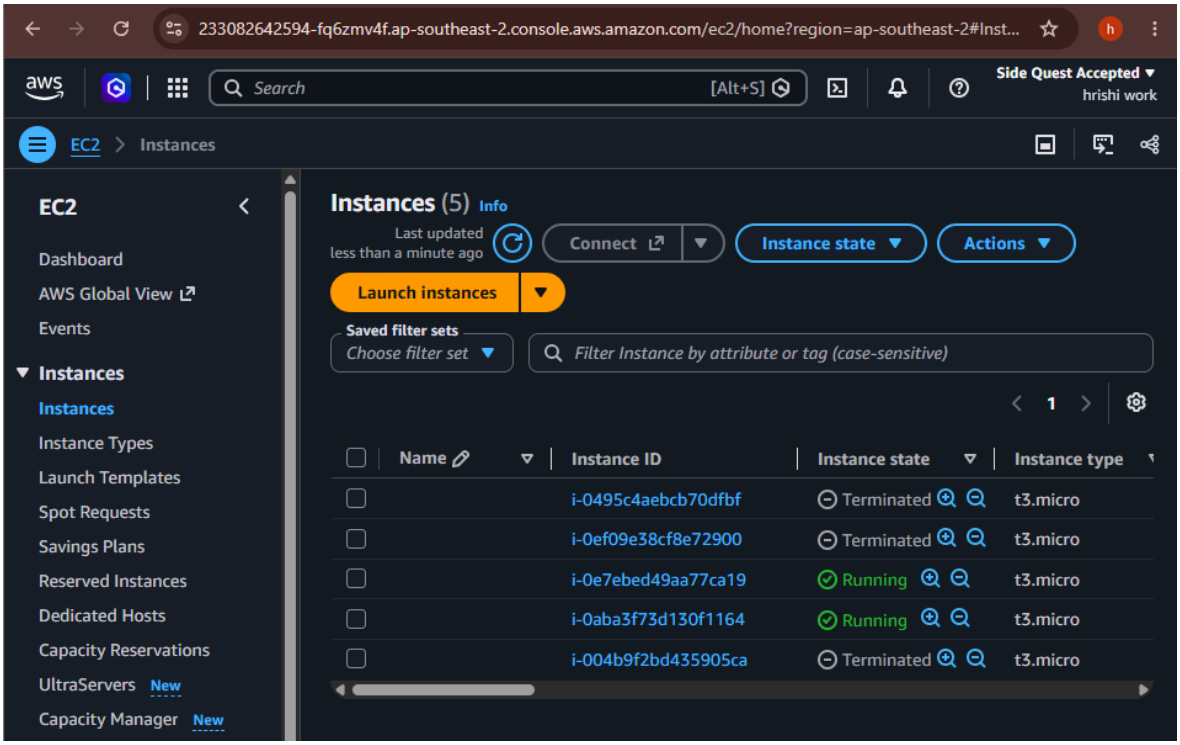

Auto Scaling groups
ALB-Web-ASG
ALB-Web-ASG
ALB-Web-ASG Capacity overview
Edit
arn:aws:autoscaling:ap-southeast-2:233082642594:autoScalingGroup:40c0045a-9a7c-4807-83c7-6c0fb536ac2a:autoScalingGroupName/ALB-Web-ASG
Desired capacity
4
Scaling limits
2 - 5
Desired capacity type
Units (number of instances)
Status
Updating capacity
Date created
Fri Oct 30 2026 15:35:58 GMT+0530 (India Standard Time)


Search
[Alt+S]
Side Quest Accepted
hrishi work
EC2 > Instances
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Instances (5) Info
Last updated less than a minute ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Filter Instance by attribute or tag (case-sensitive)
1
Name
Instance ID
Instance state
Instance type
i-0495c4aebcb70dfbf Terminated t3.micro
i-0ef09e38cf8e72900 Running t3.micro
i-0e7ebed49aa77ca19 Running t3.micro
i-0aba3f73d130f1164 Running t3.micro
i-004b9f2bd435905ca Running t3.micro


233082642594-fq6zmv4f.ap-southeast-2.console.aws.amazon.com/ec2/home?region=ap-southeast-2#Tar...
Search
[Alt+S]
Side Quest Accepted
hrishi work
EC2 > Target groups > ALB-Web-TG
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
ALB-Web-TG
Actions
Details
arn:aws:elasticloadbalancing:ap-southeast-2:233082642594:targetgroup/ALB-Web-TG/c872fb8897
Target type
Instance
Protocol : Port
HTTP: 80
Protocol version
HTTP1
VPC
vpc-05a2429f11c140618
IP address type
IPv4
Load balancer
ALB-Web-LB
4
Total targets
4
Healthy
0 Anomalous
0
Unhealthy
0
Unused
0
Initial
0
Draining
Distribution of targets by Availability Zone (AZ)
Select values in this table to see corresponding filters applied to the Registered targets table below.
Targets
Monitoring
Health checks
Attributes
Tags

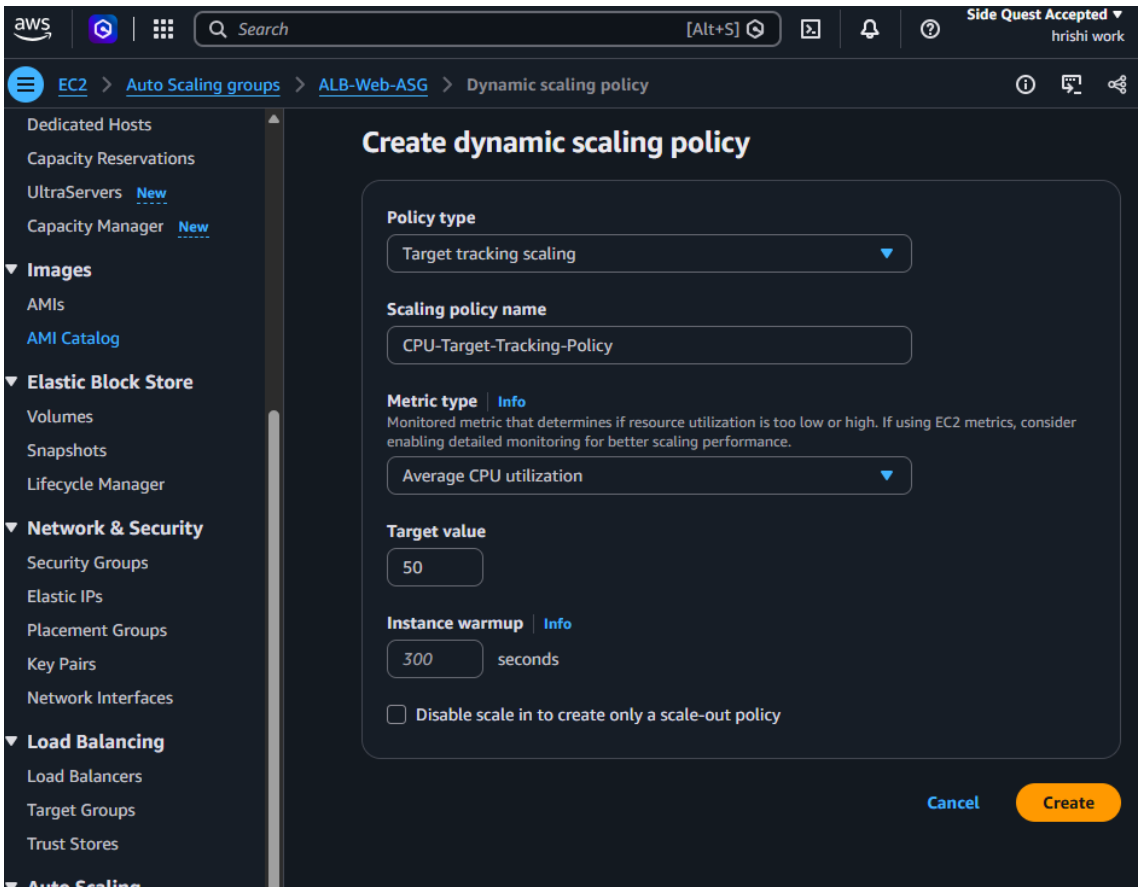

ALB-Web-ASG
ALB-Web-ASG Capacity overview
Edit
arn:aws:autoscaling:ap-southeast-2:233082642594:autoScalingGroup:40c0045a-9a7c-4807-83c7-6c0fb536ac2a:autoScalingGroupName/ALB-Web-ASG
Desired capacity
2
Scaling limits
2 - 5
Desired capacity type
Units (number of instances)
Status
Updating capacity
Date created
Fri Oct 30 2026 15:35:58 GMT+0530 (India Standard Time)

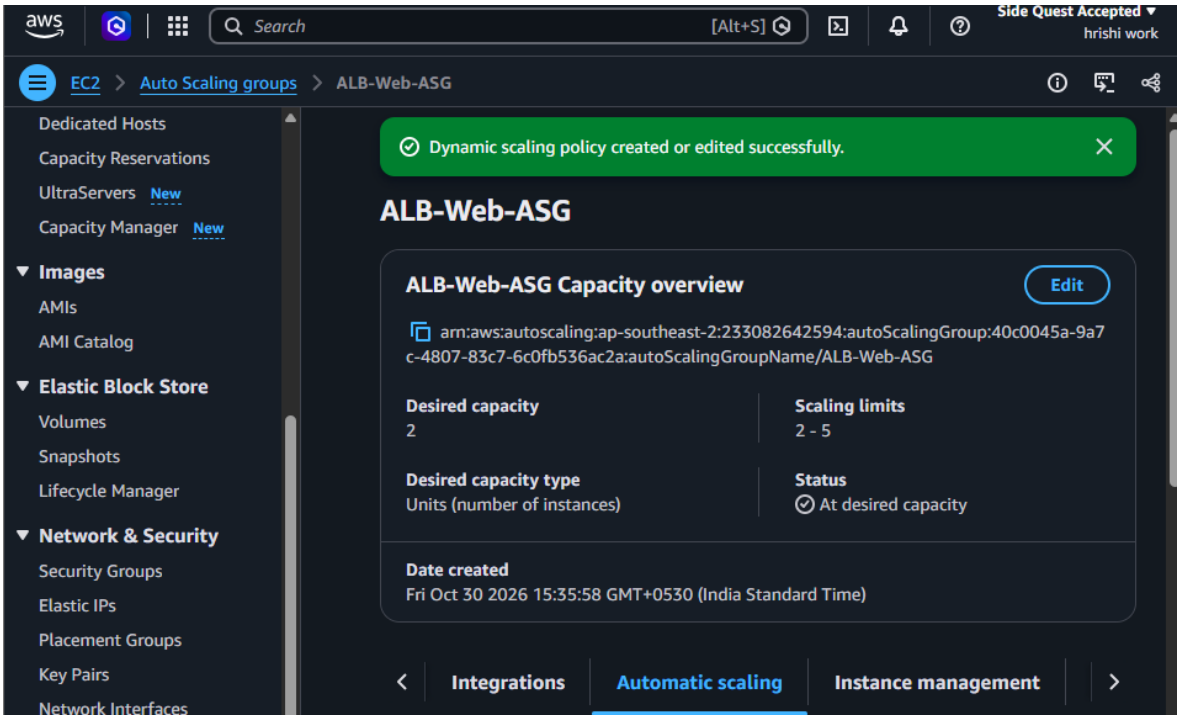

aws
Search
[Alt+S]
Side Quest Accepted
EC2
Target groups
ALB-Web-TG
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
ALB-Web-TG
Actions
Details
arn:aws:elasticloadbalancing:ap-southeast-2:233082642594:targetgroup/ALB-Web-TG/c872fb889
Target type
Instance
Protocol : Port
HTTP: 80
Protocol version
HTTP1
VPC
vpc-05a2429f11c140618
IP address type
IPv4
Load balancer
ALB-Web-LB
4
Total targets
2
Healthy
0 Anomalous
0
Unhealthy
0
Unused
0
Initial
2
Draining
Distribution of targets by Availability Zone (AZ)
Select values in this table to see corresponding filters applied to the Registered targets table below.


233082642594-fq6zmv4f.ap-southeast-2.console.aws.amazon.com/ec2/home?region=ap-southeast-2#Inst...
Search
[Alt+S]
Side Quest Accepted
hrishi work
EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Instances (5) Info
Last updated less than a minute ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Filter Instance by attribute or tag (case-sensitive)
1
Name
Instance ID
Instance state
Instance type
i-0495c4aebcb70dfbf Terminated t3.micro
i-0ef09e38cf8e72900 Terminated t3.micro
i-0e7ebed49aa77ca19 Running t3.micro
i-0aba3f73d130f1164 Running t3.micro
i-004b9f2bd435905ca Terminated t3.micro

Task 6: Dynamic Scaling Policy

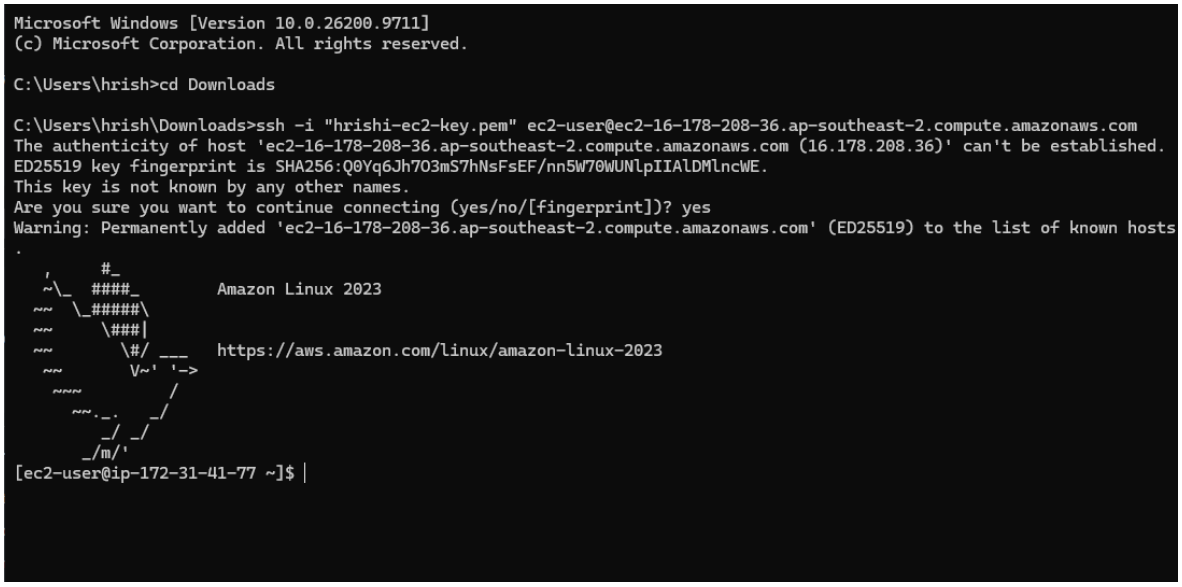
- Created target tracking policy *CPU-Target-Tracking-Policy* on *ALB-Web-ASG* with Average CPU Utilization target of 50%.
- The policy was created successfully and is active on the Auto Scaling Group.

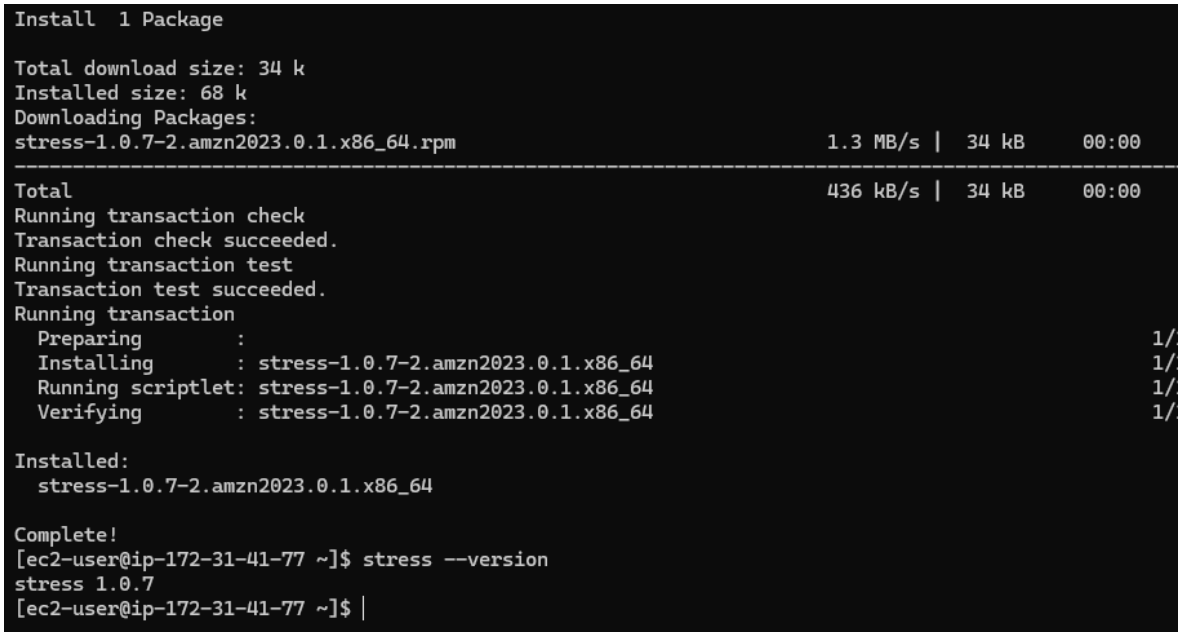


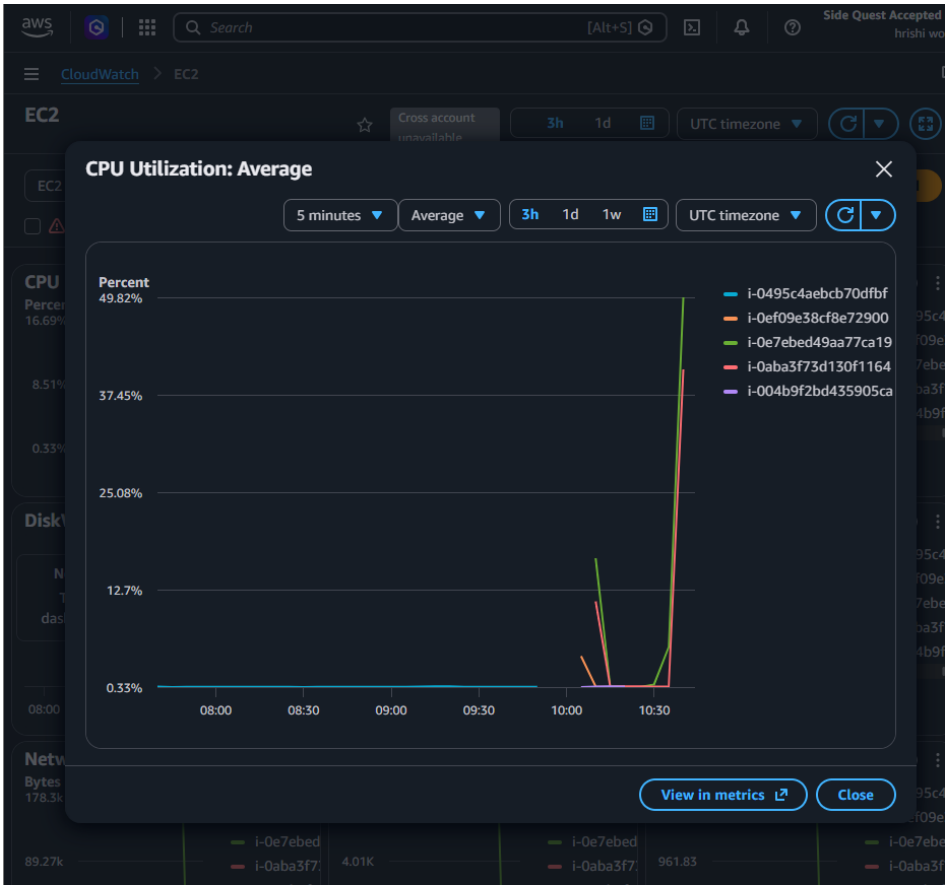


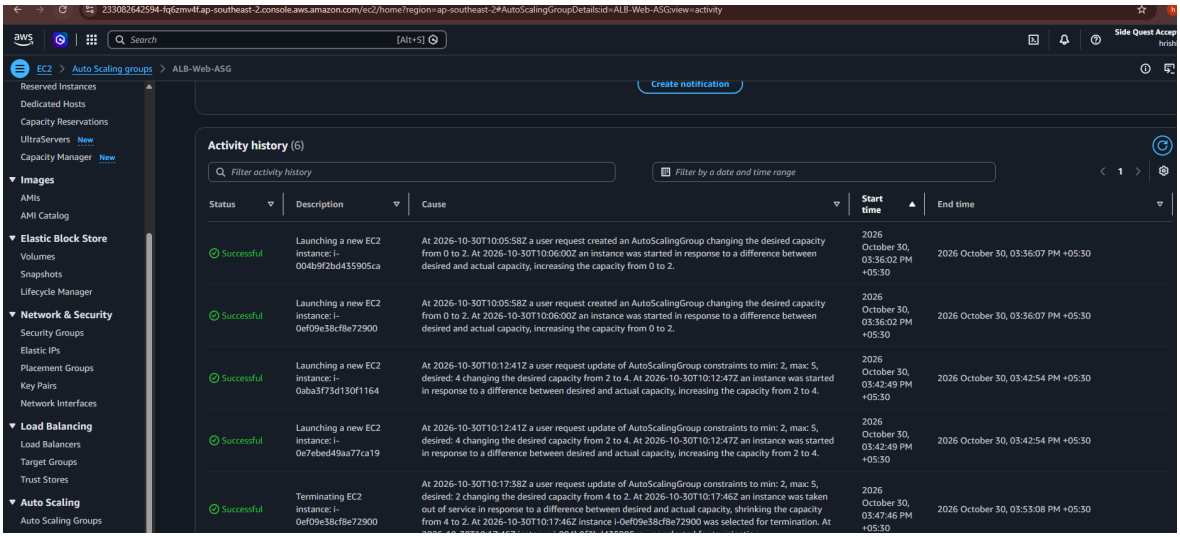
Task 7: Generate CPU Load

- Connected to an instance over SSH and installed the **stress** tool.
- Monitored CPU utilization in CloudWatch and the scaling events in the Auto Scaling activity history.






CPU Utilization: Average
5 minutes
Average
3h
1d
1w
UTC timezone
Percent
49.82%
37.45%
25.08%
12.7%
0.33%
08:00
08:30
09:00
09:30
10:00
10:30
i-0495c4aebcb70dfbf
i-0ef09e38cf8e72900
i-0e7ebed49aa77ca19
i-0aba3f73d130f1164
i-004b9f2bd435905ca
View in metrics
Close

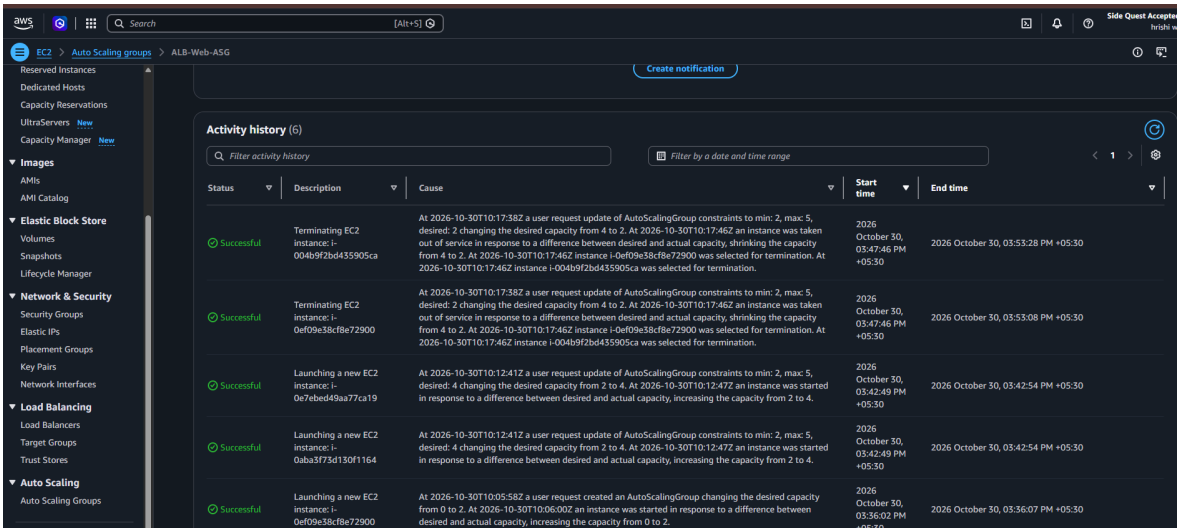

Auto Scaling groups
ALB-Web-ASG
Create notification
Activity history (6)
Filter activity history
Filter by a date and time range
Status
Description
Cause
Start time
End time
Launching a new EC2 instance: i-004b9f2bd435905ca
At 2026-10-30T10:05:58Z a user request created an AutoScalingGroup changing the desired capacity from 0 to 2. At 2026-10-30T10:06:00Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 0 to 2.
Launching a new EC2 instance: i-0ef09e38cf8e72900
Launching a new EC2 instance: i-0aba3f73d130f1164
At 2026-10-30T10:12:41Z a user request update of AutoScalingGroup constraints to min: 2, max: 5, desired: 4 changing the desired capacity from 2 to 4. At 2026-10-30T10:12:47Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 2 to 4.
Launching a new EC2 instance: i-0e7ebed49aa77ca19
Terminating EC2 instance: i-0ef09e38cf8e72900
At 2026-10-30T10:17:38Z a user request update of AutoScalingGroup constraints to min: 2, max: 5, desired: 2 changing the desired capacity from 4 to 2. At 2026-10-30T10:17:46Z an instance was taken out of service in response to a difference between desired and actual capacity, shrinking the capacity from 4 to 2. At 2026-10-30T10:17:46Z instance i-0ef09e38cf8e72900 was selected for termination. At

Task 8: Automatic Scale-In

- Stopped the stress test with Ctrl+C; CPU utilization dropped back to near 0% in CloudWatch.
- The Auto Scaling Group stayed at its minimum of 2 instances, with no extra instances left over.

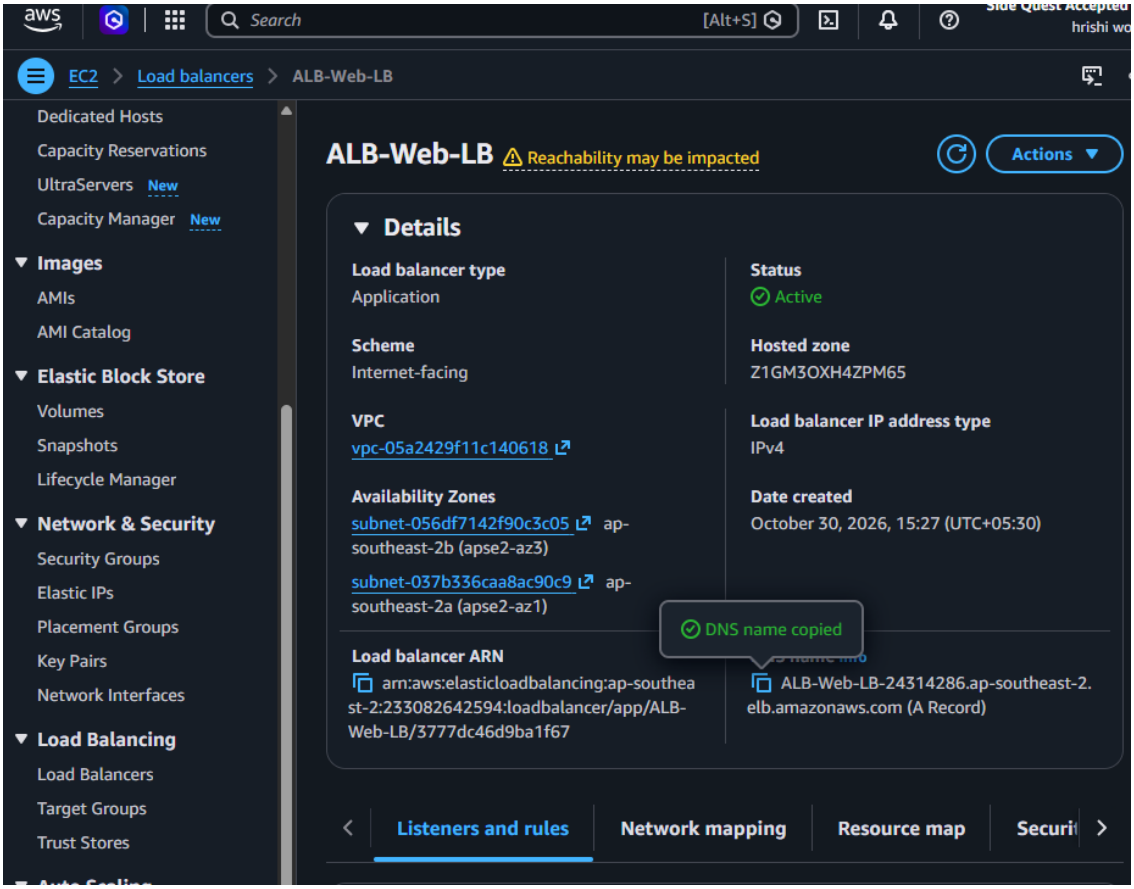
```
[ec2-user@ip-172-31-7-229 ~]$ stress --cpu 1 --timeout 600
stress: info: [27530] dispatching hogs: 1 cpu, 0 io, 0 vm, 0 hdd
^C
[ec2-user@ip-172-31-7-229 ~]$
```





Task 9: Verify Load Balancing

- Opened the ALB DNS name in the browser and sent 10 repeated requests using a PowerShell loop.
- Responses alternated between two different instances (different instance IDs and hostnames), confirming traffic is distributed.





```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\hrish> $dns = "ALB-Web-LB-24314286.ap-southeast-2.elb.amazonaws.com"
PS C:\Users\hrish>
PS C:\Users\hrish> 1..10 | ForEach-Object {
>>     Write-Host "`nRequest $_"
>>     curl.exe -s -H "Connection: close" "http://$dns"
>>     Start-Sleep -Seconds 2
>> }

Request 1
<html>
<head>
<title>Auto Scaling Lab</title>
</head>
<body>
<h1>AWS EC2 Auto Scaling Lab</h1>
<h2>Welcome to my Web Server</h2>
<p>Instance ID: i-0e7ebed49aa77ca19</p>
<p>Hostname: ip-172-31-41-77.ap-southeast-2.compute.internal</p>
</body>
</html>

Request 2
<html>
<head>
<title>Auto Scaling Lab</title>
</head>
<body>
<h1>AWS EC2 Auto Scaling Lab</h1>
<h2>Welcome to my Web Server</h2>
<p>Instance ID: i-0e7ebed49aa77ca19</p>
<p>Hostname: ip-172-31-41-77.ap-southeast-2.compute.internal</p>
</body>
</html>

Request 3
<html>
<head>
<title>Auto Scaling Lab</title>
</head>
<body>
<h1>AWS EC2 Auto Scaling Lab</h1>
<h2>Welcome to my Web Server</h2>
<p>Instance ID: i-0aba3f73d130f1164</p>
<p>Hostname: ip-172-31-7-229.ap-southeast-2.compute.internal</p>
</body>
</html>

Request 4
```

```
Request 4
<html>
<head>
<title>Auto Scaling Lab</title>
</head>
<body>
<h1>AWS EC2 Auto Scaling Lab</h1>
<h2>Welcome to my Web Server</h2>
<p>Instance ID: i-0e7ebed49aa77ca19</p>
<p>Hostname: ip-172-31-41-77.ap-southeast-2.compute.internal</p>
</body>
</html>

Request 5
<html>
<head>
<title>Auto Scaling Lab</title>
</head>
<body>
<h1>AWS EC2 Auto Scaling Lab</h1>
<h2>Welcome to my Web Server</h2>
<p>Instance ID: i-0aba3f73d130f1164</p>
<p>Hostname: ip-172-31-7-229.ap-southeast-2.compute.internal</p>
</body>
</html>

Request 6
<html>
<head>
<title>Auto Scaling Lab</title>
</head>
<body>
<h1>AWS EC2 Auto Scaling Lab</h1>
<h2>Welcome to my Web Server</h2>
<p>Instance ID: i-0e7ebed49aa77ca19</p>
<p>Hostname: ip-172-31-41-77.ap-southeast-2.compute.internal</p>
</body>
</html>

Request 7
<html>
<head>
<title>Auto Scaling Lab</title>
</head>
<body>
<h1>AWS EC2 Auto Scaling Lab</h1>
<h2>Welcome to my Web Server</h2>
<p>Instance ID: i-0aba3f73d130f1164</p>
<p>Hostname: ip-172-31-7-229.ap-southeast-2.compute.internal</p>
```

```
<html>
<head>
<title>Auto Scaling Lab</title>
</head>
<body>
<h1>AWS EC2 Auto Scaling Lab</h1>
<h2>Welcome to my Web Server</h2>
<p>Instance ID: i-0aba3f73d130f1164</p>
<p>Hostname: ip-172-31-7-229.ap-southeast-2.compute.internal</p>
</body>
</html>

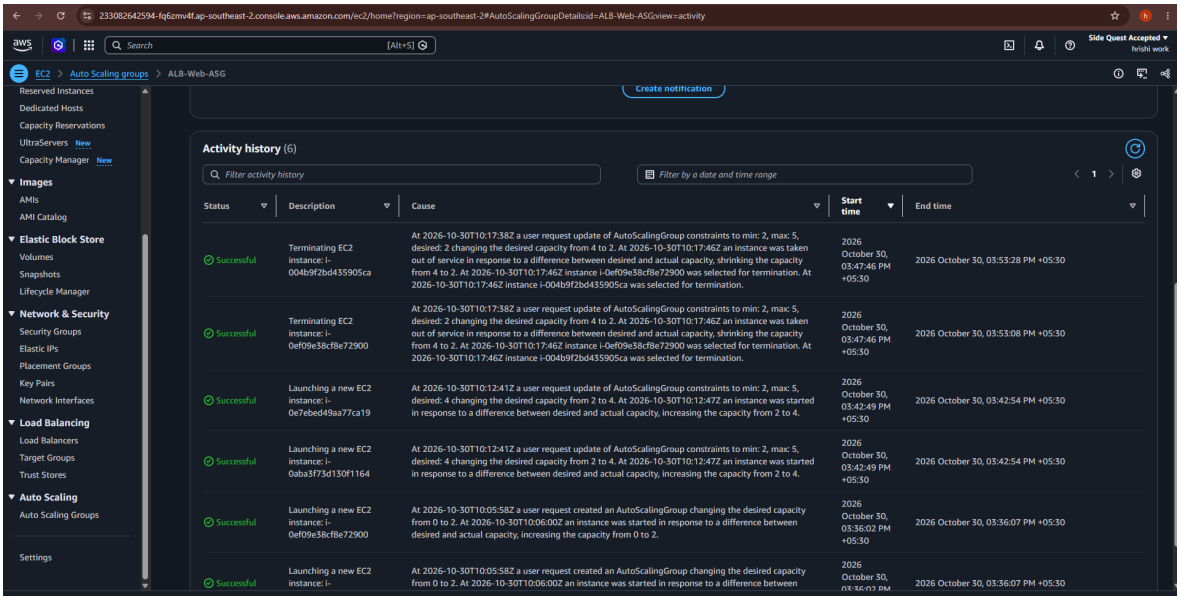
Request 8
<html>
<head>
<title>Auto Scaling Lab</title>
</head>
<body>
<h1>AWS EC2 Auto Scaling Lab</h1>
<h2>Welcome to my Web Server</h2>
<p>Instance ID: i-0aba3f73d130f1164</p>
<p>Hostname: ip-172-31-7-229.ap-southeast-2.compute.internal</p>
</body>
</html>

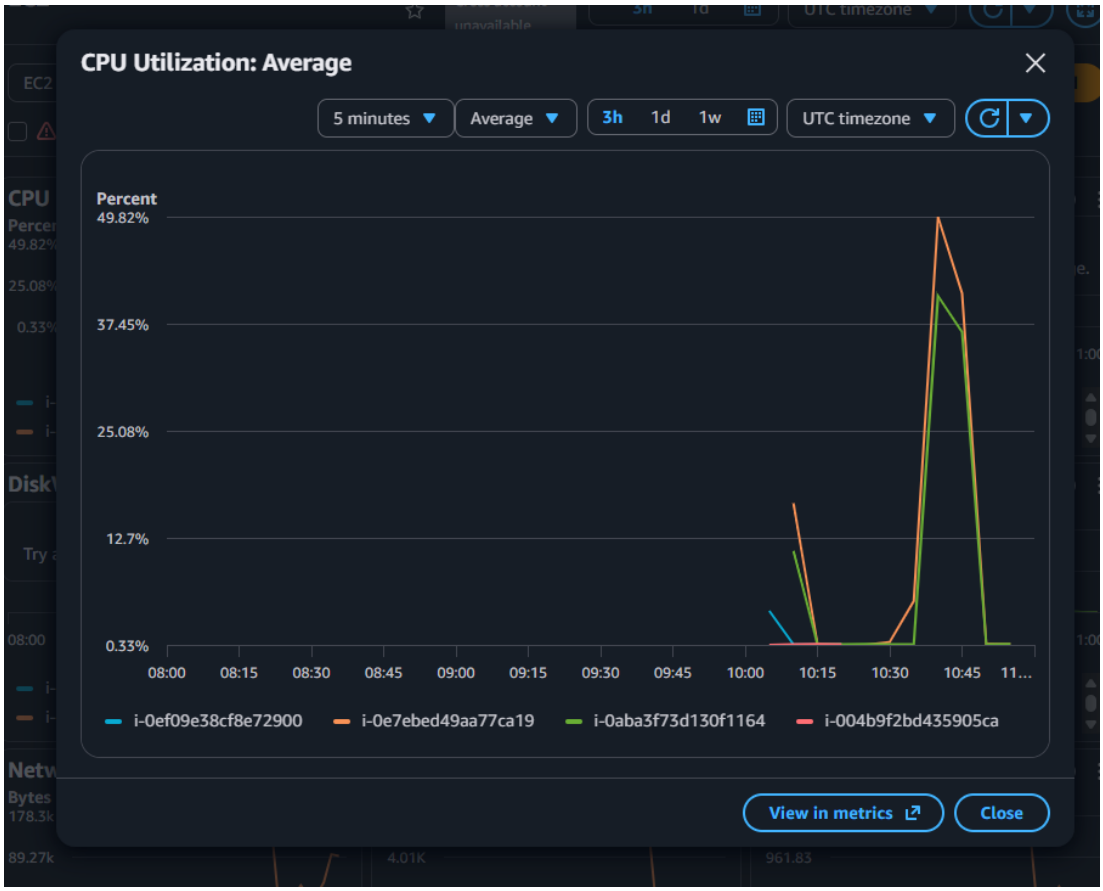
Request 9
<html>
<head>
<title>Auto Scaling Lab</title>
</head>
<body>
<h1>AWS EC2 Auto Scaling Lab</h1>
<h2>Welcome to my Web Server</h2>
<p>Instance ID: i-0e7ebed49aa77ca19</p>
<p>Hostname: ip-172-31-41-77.ap-southeast-2.compute.internal</p>
</body>
</html>

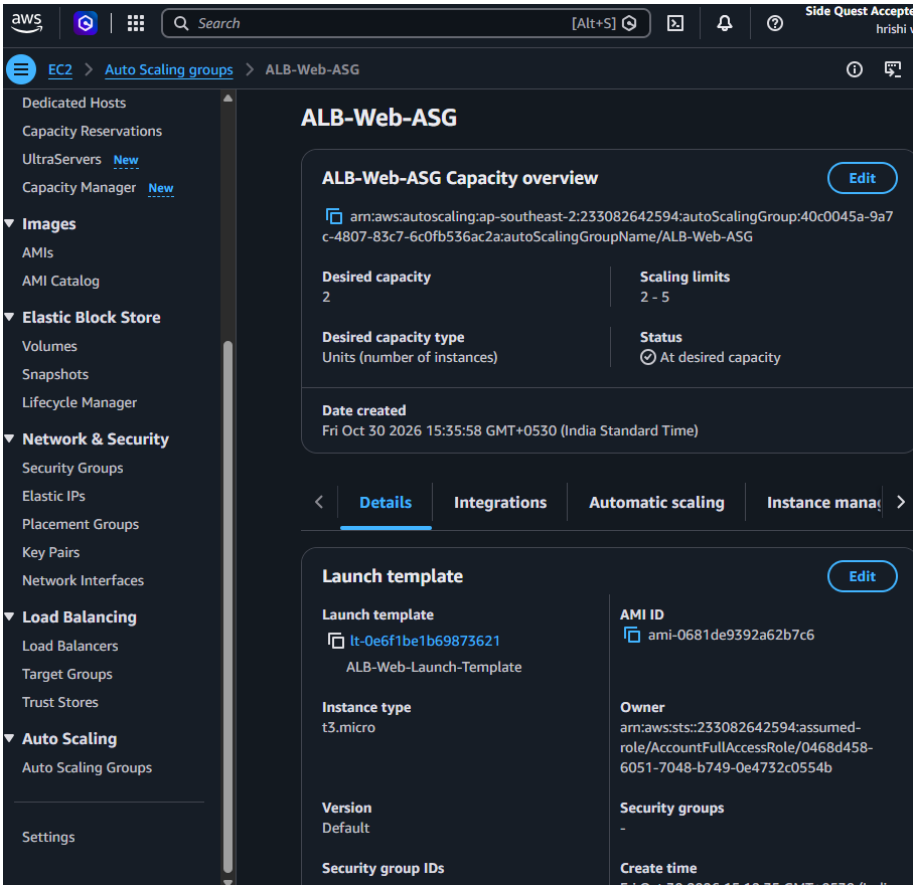
Request 10
<html>
<head>
<title>Auto Scaling Lab</title>
</head>
<body>
<h1>AWS EC2 Auto Scaling Lab</h1>
<h2>Welcome to my Web Server</h2>
<p>Instance ID: i-0aba3f73d130f1164</p>
<p>Hostname: ip-172-31-7-229.ap-southeast-2.compute.internal</p>
</body>
</html>
PS C:\Users\hrish>
```

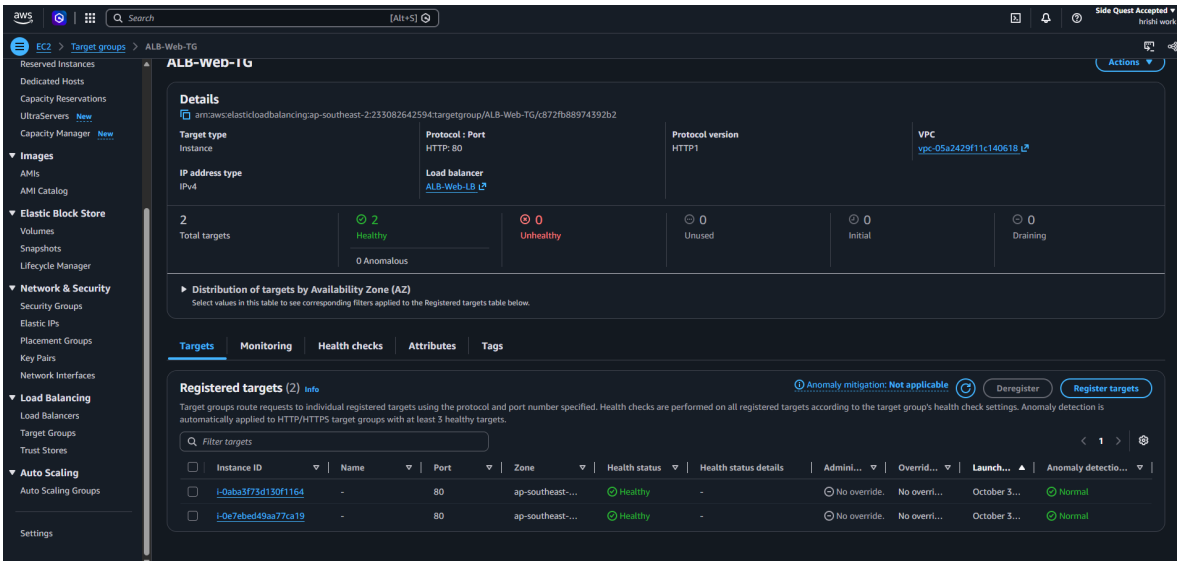
Task 10: Monitor Scaling Activities

- Reviewed the Auto Scaling activity history, which shows the launch and termination events.
- Checked CloudWatch CPU metrics, the ASG status (2 instances), and target group health (2 healthy targets).






Search
[Alt+S]
EC2 > Auto Scaling groups > ALB-Web-ASG
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
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AMI Catalog
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Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Auto Scaling Groups
Settings
ALB-Web-ASG
ALB-Web-ASG Capacity overview
Edit
arn:aws:autoscaling:ap-southeast-2:233082642594:autoScalingGroup:40c0045a-9a7c-4807-83c7-6c0fb536ac2a:autoScalingGroupName/ALB-Web-ASG
Desired capacity
2
Scaling limits
2 - 5
Desired capacity type
Units (number of instances)
Status
At desired capacity
Date created
Fri Oct 30 2026 15:35:58 GMT+0530 (India Standard Time)
Details
Integrations
Automatic scaling
Launch template
Edit
Launch template
lt-0e6f1be1b69873621
ALB-Web-Launch-Template
AMI ID
ami-0681de9392a62b7c6
Instance type
t3.micro
Owner
arn:aws:sts::233082642594:assumed-role/AccountFullAccessRole/0468d458-6051-7048-b749-0e4732c0554b
Version
Default
Security groups
-
Security group IDs
Create time


EC2 > Target groups > ALB-Web-TG
ALB-Web-TG
Actions
Details
arn:aws:elasticloadbalancing:ap-southeast-2:233082642594:targetgroup/ALB-Web-TG/c872fb88974392b2
Target type
Instance
Protocol : Port
HTTP: 80
Protocol version
HTTP1
VPC
vpc-05a2429f11c140618
IP address type
IPv4
Load balancer
ALB-Web-LB
2
Total targets
2
Healthy
0 Anomalous
0
Unhealthy
0
Unused
0
Initial
0
Draining
Distribution of targets by Availability Zone (AZ)
Select values in this table to see corresponding filters applied to the Registered targets table below.
Targets
Monitoring
Health checks
Attributes
Tags
Registered targets (2) Info
Anomaly mitigation: Not applicable
Deregister
Register targets
Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.
Filter targets
Instance ID
Name
Port
Zone
Health status
Health status details
Admini...
Overrid...
Launch...
Anomaly detectio...
i-0aba3f73d130f1164
80
ap-southeast-...
Healthy
No override.
No overri...
October 3...
Normal
i-0e7ebed49aa77ca19
80
ap-southeast-...
Healthy
No override.
No overri...
October 3...
Normal